

Freight Rate Prediction Report

1. Data Split and Validation Approach

The labelled data (data/train_test.csv, 48 000 loads) covers 2025-01-01 through 2025-10-31. The task is to predict 12000 unseen loads in data/validation.csv, which cover 2025-11-01 through 2025-12-31. This makes the problem a forecasting task, so the validation strategy was designed to mimic that:

- Train fold: 2025-01-01 to 2025-08-31 (38 477 rows)
- Holdout fold: 2025-09-01 to 2025-10-31 (9 523 rows)

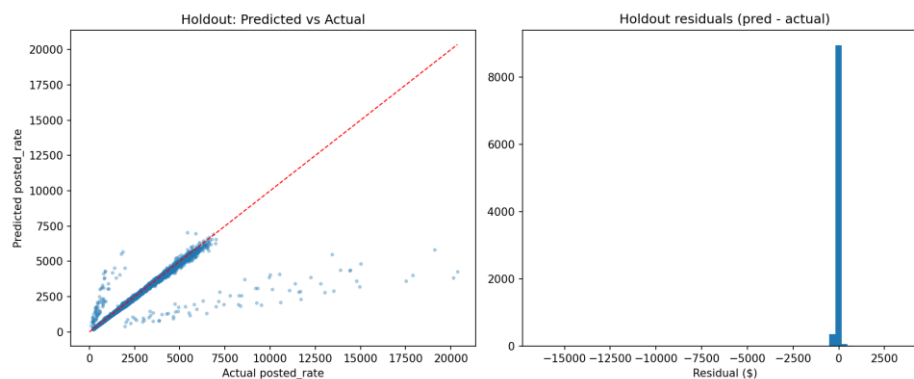
A decision was made to avoid random split because it would let the model see loads chronologically after the ones in its test fold, which overstates real world generalization and can leak lane/date patterns across the split. A chronological holdout instead reproduces the actual task by predicting rates for time periods the model has not seen.

Once the holdout confirmed the model generalizes forward in time, the final production model was refit on all 48 000 labelled rows before scoring validation.csv, since there is no reason to withhold September/October data from the model that is shipped.

Holdout performance

Metric	Value	Interpretation
MAE	\$113.33	Typical prediction is within ~\$113 of the actual posted rate
RMSE	\$636.73	Inflated by a small number of unexplained rate anomalies (Section 2)
MAPE	4.92%	Median-relative error across all holdout loads
R ²	0.826	83% of variance in posted_rate explained on unseen Sep–Oct data

Predicted vs actual rate (left) and residual distribution(right)



2. Key Data Quality Findings

- Negative weights: 292 rows (0.6%) had negative weight value with the same magnitude as valid weights, not a distinct population. Fixed with `abs()`, not dropped.
- Missing values: weight was missing in 300 train rows/165 validation rows, market index was missing 374 train rows/249 validation rows. Both imputed using medians computed on the training fold only, to avoid leaking evaluation fold information into the imputation statistic.
- Rate per mile anomalies: 1.3% of rows have posted rate/distance far outside the normal \$0.9-\$5.00/mi band that covers the other 98.7% of loads. These anomalies are spread evenly across equipment type, lane, and month, no available feature explains them, which points to an unobserved factor not captured in the provided columns. Excluding them from training only reduced holdout MAE from \$161 to \$113 and MAPE from 7.55% to 4.92% confirming they were distorting the fit from the other 98.7% of normal loads rather than teaching the model anything generalisable.
- Geocoding consistency: latitude/longitude values are fixed per city name with no conflicting entries, no cleanup needed there.

3. Model Choice

XGBoost gradient-Boosted trees.

- The feature set mixes continuous and categorical variables, and the target's relationship to distance is non linear and interacts with equipment type and lane, a good fit for tree ensembles over a linear model.
- Distance and $\log(\text{distance})$ dominate feature importance, consistent with the raw correlation and matching freight market intuition rate is overwhelmingly a function of haul length, with equipment, lane, and the market_index/quote_signal external signals contributing smaller adjustments.

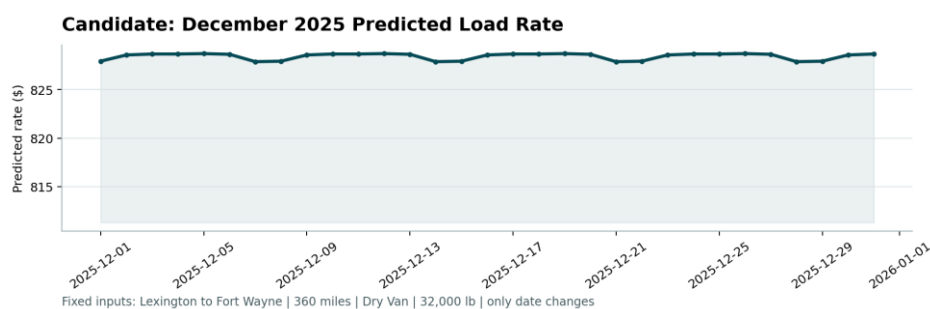
- Market_index and quote_signal are lane/row specific external signals and real values for them are present for November/December in validation.csv. This is the main channel through which the model can reflect actual future market conditions rather than purely extrapolating a training period trend.
- Requires no feature scaling, tolerates the residual missingness gracefully alongside the explicit imputation step, and trains in seconds on 48k rows, leaving room to iterate on features/anomaly handling.

4. Training and Validation Walkthrough

- src/features.py: shared cleaning (sign-fix, date parsing), leakage-safe imputation and label-encoding, and the feature set (distance, log_distance, weight, market_index, quote_signal, cyclical month, day-of-week, day-of-year, equipment/pickup/delivery codes).
- src/train.py: chronological Jan–Aug / Sep–Oct split; fits imputation and encoding on the train fold only; drops rate-per-mile anomalies from training only; trains XGBoost (600 trees, depth 6, lr 0.03, subsample/colsample 0.8); reports MAE/RMSE/MAPE/R² and saves diagnostic plots on the untouched holdout; then refits a final model on all labeled data (anomalies excluded) and saves it to outputs/model.joblib.
- src/predict.py : loads the saved model and preprocessors, applies the identical cleaning/feature pipeline to data/validation.csv, and writes outputs/validation_predictions.csv with exactly load_id,predicted_rate.

5. December Prediction Chart.

data/december_chart_inputs.csv holds pickup, delivery, distance, equipment, and weight identical (Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb) across all 31 December 2025 rows, only date changes. This isolates the model's learned date/seasonality effect from every other variable that varies elsewhere in this project. The file has no market_index/quote_signal columns, so both were filled with a single fixed value, the median historically observed on this exact lane and equipment in train_test.csv (21 matching rows), keeping every row identical except date, consistent with the file's own design. Predictions were produced by src/december_chart.py and validated by the assessment's own src/score.py, which generated the chart below directly.



The line is nearly flat (\$827–\$829) with a subtle repeating dip roughly every 7 days. This is expected given the feature set: with distance, equipment, weight, and lane held constant, month is constant too (all rows are December, so month_sin/month_cos carry zero variance within this chart), and day_of_year has very low importance (~0.2%). What's left to vary row-to-row is essentially day_of_week, so this chart is really showing the model's isolated day-of-week effect, not a broader December trend. That's a useful diagnostic in itself: it confirms the model isn't inventing spurious month-level swings when the only thing that should be changing is which day of the week a fixed lane ship on.